

ANUP PRASAD GUPTA

Staff Software Engineer | Technical Lead | Java & Spring Boot | Cloud (GCP) | Microservices | Distributed Systems | Enterprise Architecture

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PROFESSIONAL SUMMARY

Staff Software Engineer and Technical Lead with 10 years of experience designing, modernizing, and delivering mission-critical enterprise platforms — including product catalog, search & retrieval, and API systems — across healthcare, pharmacy, retail, e-commerce, payments, and subscription services for Fortune 500 organizations. Currently serving as a Staff Software Engineer / Technical Lead at CVS Health. Deep expertise across the full stack of enterprise engineering: Java 21, Spring Boot, Spring WebFlux/Reactive, Microservices, GraphQL/REST and BFF API design, Elasticsearch-based search & retrieval, PostgreSQL pgvector semantic/vector retrieval, GCP cloud platform, Kubernetes-based infrastructure, distributed low-latency systems (6,000+ requests/second), event-driven architecture, resiliency, and legacy modernization. Proven record leading onshore/offshore engineering teams, owning solution architecture end-to-end, authoring high-level/low-level design documentation, and building reusable enterprise platforms adopted across multiple teams. Strong technical leadership: mentors engineers, drives cross-team architecture alignment, serves as a technical escalation point for complex production and architecture decisions, and translates ambiguous business goals into actionable technical roadmaps. Industry- and technology-agnostic engineering leadership, with hands-on delivery spanning cloud migrations, high-throughput caching, catalog & product search systems, API gateways, CI/CD modernization, and zero-downtime production cutovers at scale (1M+ transactions/day, 10,000+ retail locations).

KEY CLIENTS & ENGAGEMENTS

- CVS Health — Fortune 500 list #6 healthcare, pharmacy, and retail enterprise; full-time Staff Engineer / Technical Lead across immunization, payments, and subscription platforms.
- BJ's Wholesale Club — Fortune 500 list #204 retail and B2B/B2C e-commerce program, delivered via Tata Consultancy Services.

STAFF-LEVEL IMPACT

- Led HCL Commerce modernization for a Fortune 500 B2B/B2C e-commerce platform using Java, Spring Boot, Microservices, AWS, HCL Commerce, and IBM API Connect; delivered the migrated website in 8 months.
- Led CVS immunization platform modernization, migrating legacy RxConnect/Oracle capabilities to a resilient GCP Spring Boot Microservices platform with zero downtime — supporting 1M+ immunizations/day and 6,000+ requests/second.
- Designed 100+ GraphQL capabilities aligned with FHIR standards and personally delivered 45+ capabilities, including solutions integrating with 12–15 external systems.
- Provided technical leadership for a 500K+ line-of-code platform with zero critical/high vulnerabilities.

CORE TECHNOLOGY

Languages & Frameworks	Java 8/17/21, Spring Boot, Spring Cloud, Spring Cloud Gateway, Spring WebFlux (Reactive/Async), Microservices, BFF (Backend-for-Frontend), Hibernate, Struts 1.x, JSP
API & Integration	GraphQL, REST APIs, IBM API Connect, GatewayScript, REST Assured, TestNG, FHIR, Swagger/OpenAPI, API Versioning, OAuth 2.0, JWT, Session Management, API Security
Cloud & Platform	Google Cloud Platform (GCP), AWS, GKE, Cloud Spanner, BigQuery, Pub/Sub, Dataflow, Cloud Composer (Airflow), AWS CloudWatch
Messaging & Caching	Kafka (Event Producer/Consumer), Redis, Event-Driven Architecture, Asynchronous Processing, Retry/Recovery Patterns
Databases	Oracle, PostgreSQL (incl. pgvector Vector Search), Cloud Spanner, MongoDB, MySQL, DynamoDB, DB2, RDS
Search, Data & Monitoring	Elasticsearch, BigQuery, SQL, Vector/Semantic Search, Kibana, NewRelic, Scalyr, Splunk, Database & Data Validation, Enterprise Logging Frameworks
Search & Information Retrieval	Elasticsearch-based Search Indexing & Retrieval, GraphQL/REST Catalog & Product Search, PostgreSQL pgvector (Vector/Semantic Retrieval), Query Performance Optimization, Distributed Low-Latency Systems
Commerce	HCL Commerce, IBM WebSphere Commerce (WCS), B2B/B2C E-commerce
CI/CD & Container Platforms	GitHub Actions (GHA), Jenkins, ArgoCD/GitOps, Kubernetes (K8s), k3s, Docker, Bamboo, Automation Suites
Networking & Load Balancing	F5 GTM/LTM, Global & Local Load Balancing, Traffic Management, High Availability Routing
Tools & Platform Engineering	IntelliJ IDEA, Spring Tool Suite, Eclipse, RAD, Postman, JMeter, SOAP UI, JIRA, Bitbucket, Confluence, Git, Internal Spring_INITIALIZER/Scaffolding Platforms

Engineering & Leadership	Distributed Systems, Enterprise Architecture, Cloud Migration, Legacy Modernization, Resiliency, High Availability, Performance Engineering, SRE, Production Readiness, Technical Mentoring, Agile, Client & Stakeholder Management
Quality, Security & Compliance	Unit Testing, Code Reviews, Root-Cause Analysis, Defect Prevention, Snyk Vulnerability Remediation, ADA/PCI/PII Compliance Awareness
Design & Documentation	HLD/LLD Design Documents, System/Request Flow Diagrams, API & Architecture Documentation, Confluence

PROFESSIONAL EXPERIENCE

CVS HEALTH | Staff Software Development Engineer | Technical Lead

April 2025 – Present | Richardson, TX

- Own technical strategy, solution architecture, technical design, and delivery across distributed onshore and offshore engineering teams.
- Translate complex business and functional requirements into scalable technical solutions; decompose initiatives, assign work, remove blockers, and drive high-quality delivery.
- Own architecture decisions across service boundaries, frameworks, databases, integration patterns, scalability, resiliency, performance, security, and maintainability.
- Lead POCs and technical evaluations for framework, database, integration, scalability, performance, and operational choices before implementation.
- Establish reusable enterprise engineering patterns and standards and influence technical decisions across teams and applications.
- Provide architecture/design reviews, code reviews, technical mentoring, risk identification, dependency management, and production-readiness leadership.
- Designed and built an enterprise-wide Async Engine standardizing external fire-and-forget and asynchronous integrations with retry and failure-handling mechanisms.
- Own and scale the enterprise Spring Cloud Gateway platform (Domain Gateway), the standardized microservices entry point with configuration-driven, no-code onboarding across CVS domains.
- Own and scale the enterprise Kafka Event Producer platform (Event Publisher), standardizing event-driven integrations and event publishing across teams.
- Identify application-specific solutions that can become reusable enterprise platforms, reducing duplicated engineering effort and improving consistency.
- Serve as a technical escalation point for complex architecture, integration, performance, and production issues.
- Built CVS's own internal Spring Initializer platform, a self-service scaffolding tool letting teams bootstrap production-ready domain applications with pre-wired connectivity to Oracle, GCP Cloud Spanner, PostgreSQL, Redis, and Kafka event producer/consumer patterns.
- Adopted PostgreSQL pgvector for vector/semantic search use cases and Spring WebFlux/Reactive programming to build non-blocking, asynchronous BFF (Backend-for-Frontend) services aggregating downstream microservice calls.
- Designed and built an enterprise-wide logging and observability framework, and an enterprise-wide GCP data verification framework covering store-level data validation, sales reporting, and log analysis using BigQuery.
- Modernized CI/CD tooling by introducing GitHub Actions (GHA) alongside Jenkins and adopting ArgoCD/GitOps deployments; standardized k3s for GCP workloads and Kubernetes (K8s) for edge/in-store infrastructure.
- Directed F5 GTM/LTM based global and local load-balancing and traffic-management strategy for high-availability production routing.
- Led remediation of vulnerabilities identified via Snyk scanning across enterprise services, driving the platform to zero critical/high findings; used GCP Cloud Composer (Airflow) to orchestrate data pipelines and automation workflows.
- Author high-level and low-level design (HLD/LLD) documents and system-to-system request-flow diagrams, and drive design documentation and knowledge management in Confluence.
- Mentor engineers on system design and coding best practices, and drive cross-team technical alignment, setting engineering standards adopted enterprise-wide.

Technology: Java 21, Spring Boot, Spring Cloud, Spring WebFlux, Microservices, GCP, AWS, GKE, GraphQL, FHIR, Kafka, Redis, BigQuery, Cloud Spanner, Cloud Composer, Pub/Sub, Dataflow, Oracle, PostgreSQL (pgvector), MongoDB, REST APIs, Hibernate, Reactive Programming, GitHub Actions, Jenkins, ArgoCD, Kubernetes, k3s, Snyk, F5 GTM/LTM

CVS HEALTH | Senior Software Engineer II

September 2022 – April 2025 | United States

- Developed Java/Spring Boot Microservices and GraphQL/REST APIs for high-scale healthcare, pharmacy, payment, subscription, and retail platforms.
- Delivered 45+ GraphQL capabilities aligned with FHIR, including complex capabilities requiring 12–15 external integrations.

- Engineered resilient GCP payment processing with retry, fallback, and recovery mechanisms to maintain continuity during bank and Worldpay outages; completed migration with zero downtime.
- Migrated ExtraCare subscription services to GCP by developing cloud-native services and integrations.
- Built Spring Boot + Redis caching platform supporting approximately 5,000 average TPS with scheduled cache refresh.
- Built an enterprise Async Engine for fire-and-forget/asynchronous integrations with retry and failure handling.
- Created data verification capabilities using BigQuery and databases for validation, debugging, metrics, and health checks.
- Developed EDGE JVM cache refresh supporting approximately 10,000 stores in under 5 minutes and automated production validation across approximately 10,000 stores.
- Migrated 100M+ immunization records in two stages — legacy Oracle to GCP Cloud Spanner as the central data store, then GCP Cloud Spanner to store-local PostgreSQL databases across 10,000 stores via an event-driven architecture — for fast, resilient in-store access.
- Designed BFF (Backend-for-Frontend) services aggregating GraphQL/REST calls across microservices for client applications; worked extensively across Oracle, PostgreSQL, Cloud Spanner, MongoDB, and Redis for transactional, document, and caching workloads.
- Authored design documents (HLD/LLD) and system/request-flow diagrams illustrating how requests move across services, and maintained API and architecture documentation in Confluence.
- Took ownership of production issue triage, root-cause analysis, and performance optimization as part of SRE responsibilities; remediated vulnerabilities flagged by Snyk scanning.
- Performed performance optimization, root-cause analysis, production support, code reviews, testing, and defect resolution with focus on quality, security, reliability, and maintainability.

Technology: Java 17, Spring Boot, Spring Cloud Gateway, Microservices, BFF, GCP, GKE, GraphQL, Redis, Kafka, BigQuery, Cloud Spanner, Pub/Sub, Oracle, PostgreSQL, MongoDB, REST APIs, OAuth 2.0, JWT, Hibernate, Reactive Programming, REST Assured, Snyk

TATA CONSULTANCY SERVICES (TCS) | Technical Lead (progressed from Assistant System Engineer)

September 2016 – September 2022 | Kolkata, India (2016–2020) → Onsite, Massachusetts, United States (2021–2022)

Client: BJ's Wholesale Club, Fortune 500 list #204 (B2B/B2C e-commerce; combined program team size of 72). Progressed over 6 years through Assistant System Engineer → System Engineer → IT Analyst/Senior Software Engineer → Technical Lead.

- Delivered the complete Member and Catalog subsystems for BJ's Wholesale Club (per HCL Commerce's Member/Catalog architecture), building top features including "Keep Me Signed In," Cobrand/Capital One Membership & Credit Card, virtual one-time-use credit card, the Optical platform, Product List & Product Details pages (catalog browse, search, and retrieval), and OMS capabilities (member search, member list, create member).
- Implemented Elasticsearch-based search indexing and retrieval to power fast, relevant product catalog search and browse experiences across the e-commerce storefront.
- As Technical Lead, directed 5 engineers offshore and later served as the sole onshore technical point of contact after transitioning onsite, coordinating stakeholders, clarifying requirements, and driving technical direction.
- Served 2.5+ years as IBM API Connect (APIC) Technical Lead, leading a team of 10 developers; designed and delivered 100+ APIs including API design, policies, GatewayScript, security, API versioning strategy, and Swagger documentation.
- Served 2.5+ years as REST Assured Framework Technical Lead, leading a team of 10 developers; built a REST API automation and data-driven testing framework (REST Assured, TestNG) that improved regression coverage.
- Owned end-to-end development of Order and Payment domain capabilities for BJ's Wholesale Club, from requirements through production support.
- Designed and developed Spring Cloud Gateway solutions for microservices routing and API gateway capabilities.
- Led modernization and migration of IBM WebSphere Commerce (WCS) to newer platform versions, including WCS Headless Commerce, customization, integration, testing, and deployment.
- Implemented OAuth 2.0 and JWT-based authentication/authorization, token validation, and session management across customer-facing APIs, applying OOP/MVC design patterns and multi-threaded Java programming.
- Drove CI/CD automation (Bamboo, Jenkins), Agile branching strategy (Sprint/EPIC/Release), architecture design documentation, and production monitoring/troubleshooting using Kibana, NewRelic, and Scalyr.
- Applied working knowledge of ADA, PCI, and PII regulatory compliance in API and application design; conducted defect prevention, root-cause analysis, and performance troubleshooting.
- Participated directly in client interactions and requirement-gathering sessions, independently owning multiple modules end-to-end with zero critical/high-severity vulnerabilities in delivered releases.

Technology: Java, Spring Boot, Spring Cloud Gateway, Microservices, IBM API Connect, GatewayScript, IBM WebSphere Commerce (WCS), REST APIs, REST Assured, TestNG, OAuth 2.0, JWT, Struts 1.x, AWS, MySQL, PostgreSQL, MongoDB, DynamoDB, DB2, Redis, Elasticsearch, Kafka, Hibernate, Reactive Programming, Postman, JMeter, SOAP UI, Swagger, JIRA, Bitbucket, Confluence, Bamboo, Jenkins, Git, Kibana, NewRelic, Scalyr

CERTIFICATIONS

- Google Cloud Associate Cloud Engineer

- [Software Architect \(Microservices\) — Edureka](#)
- [REST API \(Intermediate\) Certificate — HackerRank](#)
- [Software Engineer — HackerRank](#)

KEY ACHIEVEMENTS & RECOGNITION

- **Payment Resiliency — Enterprise/DDAT Recognition:** Built payment resiliency within CVS's enterprise payment platform for the retail business, ensuring customer orders continue to be accepted even when the Worldpay or Vantiv payment processor is down; recognized with enterprise-level (DDAT) recognition for safeguarding revenue continuity across CVS retail operations.
- **IVR Voice & Chat Platform — Enterprise Recognition:** Recognized for contributions to the enterprise IVR voice and chat platform at CVS Health, improving customer self-service capability at scale.
- **Immunization Platform Modernization — Enterprise Recognition:** Recognized for delivering the CVS Health immunization platform modernization — migrating the legacy RxConnect application to Google Cloud Platform using Java Spring Boot Microservices — in a record 8 months, resulting in a resilient platform serving 1M+ immunizations/day and 6,000+ requests/second.
- **Immunization Data Migration — Enterprise Recognition:** Recognized for a two-stage migration of 100M+ immunization records: (1) legacy Oracle to GCP Cloud Spanner as the central data store, and (2) GCP Cloud Spanner to store-local PostgreSQL databases across 10,000 stores via an event-driven architecture.

KEY PLATFORMS/PRODUCTS

- **CVS Inbuilt Capability Portal (powered by Spring Initializer):** Built and own this self-service platform enabling engineering teams across CVS to scaffold production-ready domain applications with built-in connectivity to Oracle, GCP Cloud Spanner, PostgreSQL, Redis, and Kafka producer/consumer patterns; adopted enterprise-wide as the standard way to bootstrap new domain services.
- **Domain Gateway (Spring Cloud Gateway):** Built this enterprise-wide API gateway platform, internally known as the Domain Gateway, providing a standardized microservices entry point with configuration-driven, no-code onboarding; adopted as the enterprise standard for domain-level request routing and security.
- **Redis Caching App:** Built this enterprise-wide Spring Boot + Redis caching product supporting up to ~5,000 average TPS with scheduled cache refresh; adopted as a shared caching capability across multiple CVS domains and teams.
- **Event Publisher (Kafka Event Producer Platform):** Built this enterprise-wide Kafka Event Producer product with configuration-driven, no-code onboarding to standardize event-driven integrations and simplify event publishing; adopted across teams as the standard way to publish domain events.

EDUCATION

Techno India University — Bachelor's Degree, Electrical, Electronics and Communications Engineering (2012 – 2016), DGPA 8.89